

Abhyuday Vaish

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EDUCATION

Vellore Institute of Technology

Bachelor of Technology in Computer Science (CGPA: 9.02)

Chennai, India

July 2024 – May 2028

Delhi Public School, Lucknow | Class X: 95.6% — Class XII: 86.8%

TECHNICAL SKILLS

Languages: Python, C/C++, Java, SQL (Postgres), HTML/CSS/JS, MATLAB

Frameworks: Flask, FastAPI, PyTorch, TensorFlow, LangChain, HuggingFace, Scikit-learn, OpenCV

Tools: Git, Docker, Cursor, Claude Code, Google Cloud Platform, Render, VS Code, Linux/Bash

Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Linear Algebra, Probability & Statistics, Computer Networks

CERTIFICATIONS

- **Deep Learning (NPTEL – IIT Madras):** Elite + Gold.
- **Machine Learning Specialization (Stanford University).**
- **Generative AI for Software Development (DeepLearning.AI).**

EXPERIENCE

Samsung PRISM Intern, Samsung R&D

March 2026 – Present | Remote

- Recast multi-sample Chain-of-Thought aggregation as a QUBO selection problem, solved with Simulated Annealing.
- Ran supervised and DPO fine-tuning on Llama-3.2-3B using QUBO-curated reasoning data.
- Final DPO alignment reached 88% GSM8K and 62.96% BBH, outperforming the official baseline.

AI/ML Development Intern, Daira EdTech

Aug 2025 – Oct 2025 | Remote

- Automated a pipeline ingesting 100k+ UPSC PYQs, using OpenCV (adaptive thresholding, CLAHE) for high-accuracy text extraction from scanned documents.
- Launched a RAG/Generative AI system transforming static questions into dynamic assessments with auto-generated explanations, with tester feedback reporting ~40% improved engagement over the previous version.

Summer Research Intern, Vellore Institute of Technology

May 2025 – July 2025 | Chennai, India

- Achieved 91.4% recall and 90.8% precision (F1: 0.911) in identifying opening repertoire vulnerabilities, detecting weaknesses at an 89.5% rate (95% CI: 87.7–91.3) across 1,118 opening lines.
- Fine-tuned a Gemma-2B LLM to generate automated counter-strategies, achieving 81.6% validation accuracy.
- Analyzed 2,500+ games through a custom pipeline to distill exploitable tactical patterns and vulnerabilities.

Research Intern, IIT BHU

Mar 2026 – Apr 2026 | Remote

- Conducted literature review on deep learning methods for biomedical applications.

PROJECTS

Solar CME Detection System | Python, Flask, TensorFlow, PyTorch, Keras, PostgreSQL

Jan 2026 – May 2026

- Developed a stacked LSTM pipeline forecasting Kp-index 1h ahead from 24h solar wind windows, benchmarked against PINN and Quantum architectures on 14,700+ samples (± 1 Kp tolerance).
- Integrated an ETL pipeline ingesting 20 months of Aditya-L1 ASPEX telemetry alongside real-time DSCOVR/NASA feeds into a PostgreSQL datastore.
- Deployed a Flask dashboard with role-based access (scientist/viewer), real-time CME shock alerts via statistical anomaly detection, and scientist-restricted on-demand model retraining.

Traffic Violation Detection System | Python, YOLOv8, PyTorch, OpenCV, ByteTrack, EasyOCR, Streamlit

June 2026

- Fine-tuned 3 custom YOLOv8 models (helmet, seatbelt, license plate) on a 6,500-image dataset, achieving 94.3% mAP@50 and 90.6% F1 averaged across tasks.
- Applied ByteTrack-based trajectory analysis and geometric heuristics to identify 6 traffic violations in real time, integrating EasyOCR, SQLite, ReportLab, and Streamlit for automated e-challan generation and analytics.

AI Academic Assistant | FastAPI, Next.js, Supabase, Playwright, Tesseract OCR, pgvector

June 2025 – May 2026

- Architected an AI academic assistant indexing 850+ PYQs across 173 courses with 5 context-aware chat modes for guided exam prep.
- Built OCR + retrieval pipelines for note summarization and syllabus-aligned quizzes across 846 previous-year papers and 799 extracted questions; containerized services and automated CI/CD via GitHub Actions.
- Automated PYQ-trend-driven question paper generation across 8 syllabus modules to help students target high-weightage topics.

AWARDS & ACHIEVEMENTS

- **1st Place**, Industry-Academia Innovators Conclave (IAIC 2025): Research Poster Presentation Category.
- **Global Rank 754** in CodeChef Starters 224 (Max Rating 1450+).
- **Max Ratings:** LeetCode (1730+), Codeforces (1330+).

POSITIONS OF RESPONSIBILITY

Project Lead | AI/ML Team Member, Microsoft Innovations Club, VIT Chennai

Sep 2024 – Present | Chennai, India

- **Project Lead (Jan 2026 – Present):** Leading a 7-member team on the “AI Academic Assistant”; conducting code reviews, overseeing deployment, and mentoring juniors on GenAI.

- **AI/ML Member:** Organizing workshops and collaborating with the AI/ML team on Kaggle competitions and technical projects.

Data Science Member, GDG VIT Chennai

Sep 2025 – Present